

Wind Speed Analysis and Wind Energy Potential Assessment

1. Introduction

Wind energy is a fundamental component of sustainable energy systems. Recognising its role in promoting environmental responsibility can motivate stakeholders to support wind power development.

The primary objective is to determine the wind characteristics at a height of 50 metres and to estimate the potential for wind energy production at the site. This analysis aims to support informed decisions that can advance renewable energy initiatives and foster sustainable development.

2. Dataset Description

The dataset comprises hourly meteorological observations collected over approximately 10 years, providing a comprehensive record of atmospheric conditions. The analysis included several key variables:

- wind speed measured at a height of 50 meters (WS50M),
- wind direction at 50 meters (WD50M),
- temperature recorded at 2 meters above ground (T2M),
- surface pressure (PS), and
- relative humidity at 2 meters (RH2M).

This high temporal resolution, with hourly data, enables a detailed examination of wind variability, capturing short-term fluctuations and long-term climate trends for comprehensive wind resource assessment.

3. Exploratory Data Analysis

An initial exploratory analysis was conducted to understand the statistical properties of the wind speed data. A histogram of wind speeds showed a positively skewed distribution, with most values falling within the moderate range of approximately 4-10 m/s, as shown in Figure 3.1.

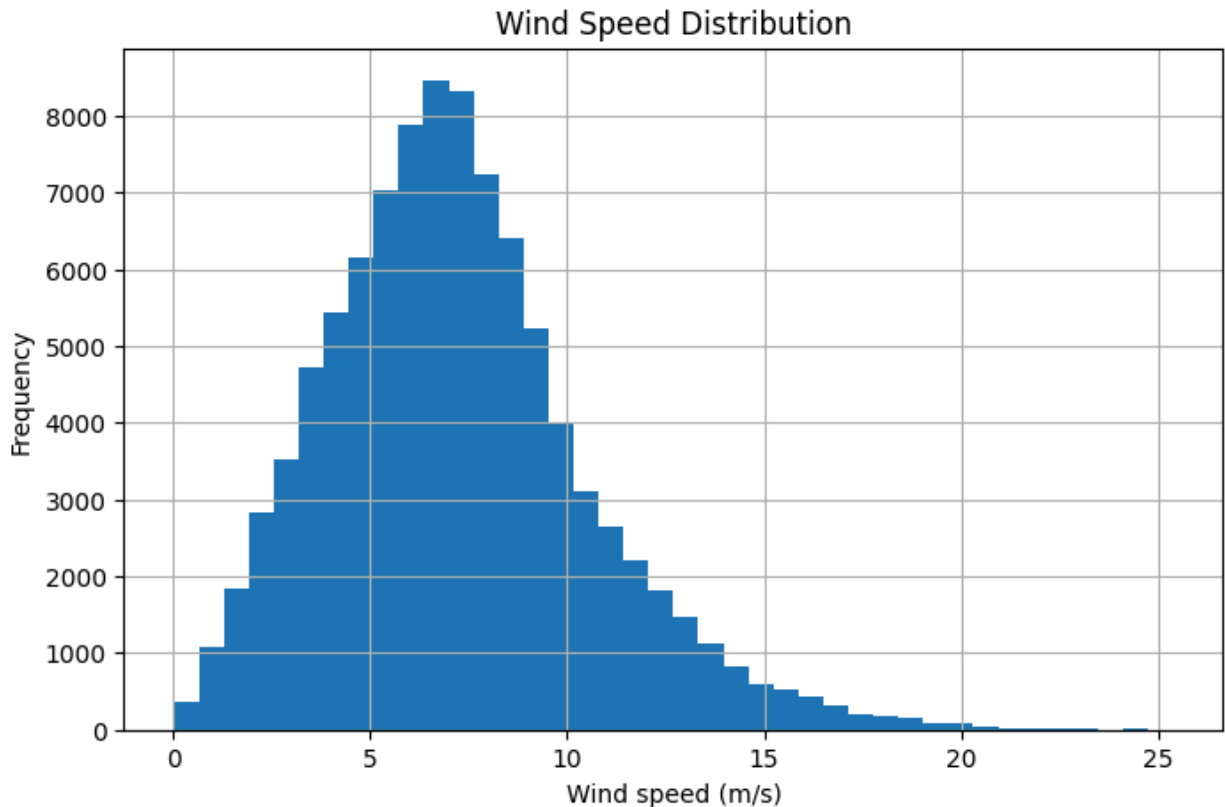


Figure 3.1 Wind Speed Distribution

The average wind speed at a height of 50 meters was **calculated to be 7.19 m/s**. This figure suggests a relatively advantageous wind regime. In numerous wind resource classification systems, mean wind speeds exceeding 7 m/s are typically regarded as suitable for wind power development.

4. Weibull Wind Speed Distribution

The Weibull distribution is commonly used to analyse wind speeds because it provides a reliable two-parameter statistical model (k for shape and c for scale) that accurately describes the probability distribution of wind speeds across various locations. This makes it easier to estimate how much wind energy could be generated.

When the Weibull distribution is fitted to actual wind speed data, the resulting curve aligns well with the observed data, indicating that the model effectively captures wind speed variability at the site. Additionally, the distribution indicates that most wind speeds fall within the operational range of modern turbines, which typically begin producing electricity at about 3-4 m/s, as illustrated in Figure 4.1.

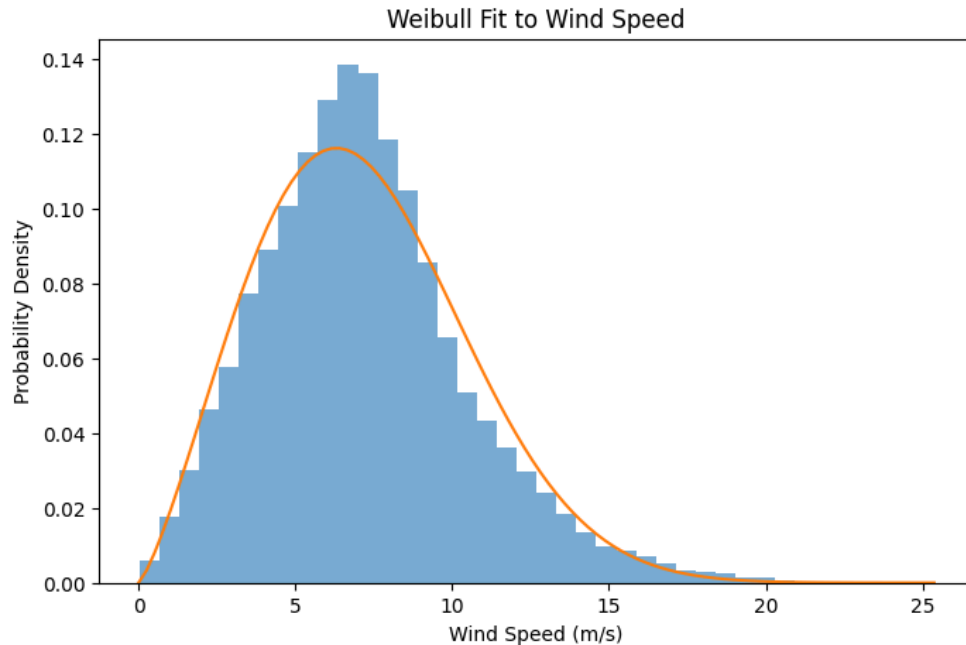


Figure 4.1 Weibull Distribution

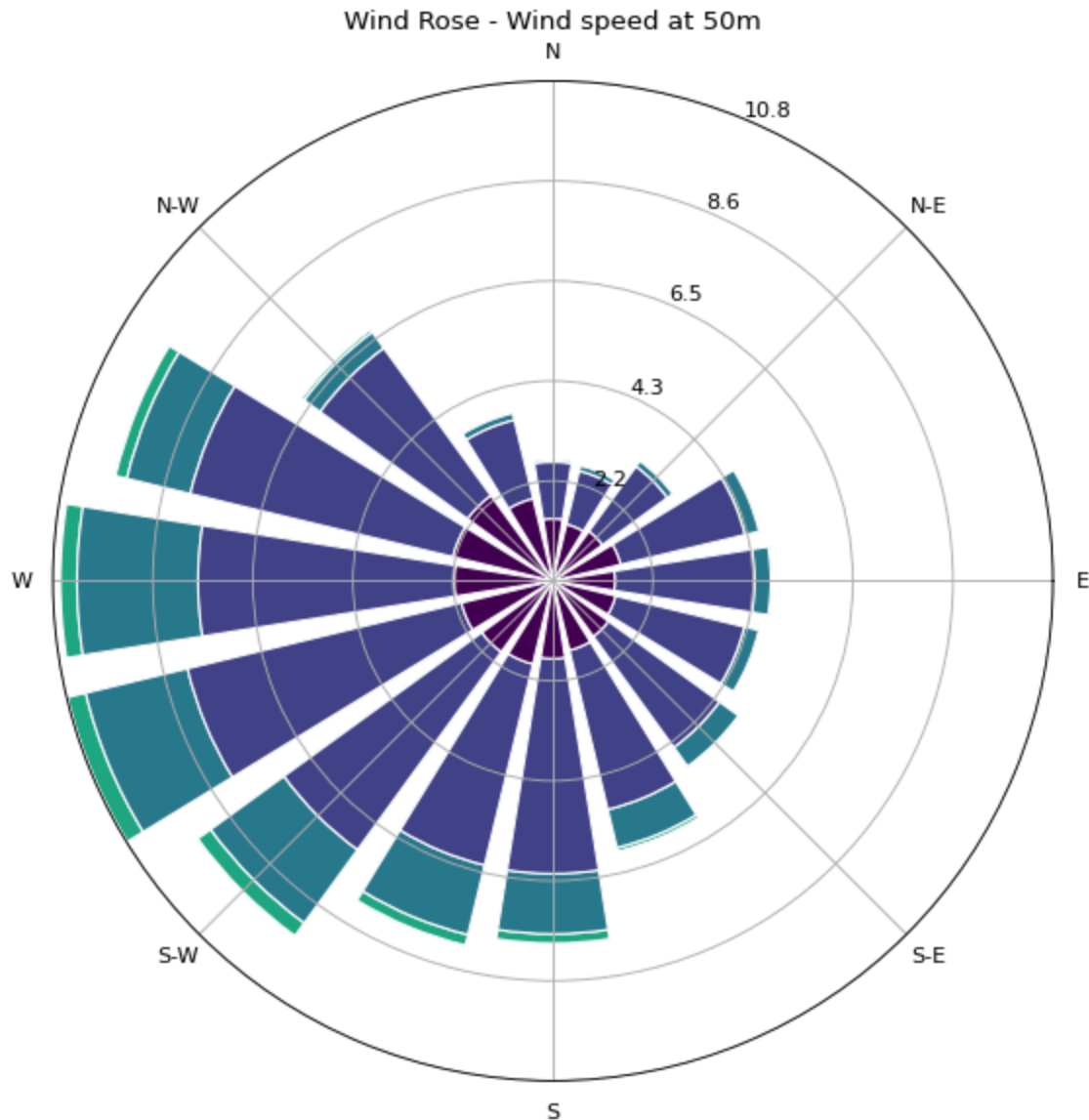
5. Wind Resource Assessment

Wind Power Density

Wind power density was estimated using the standard wind energy equation, which relates wind speed to the available kinetic energy in the airflow. The calculated mean wind power density was **388 W/m²**. According to typical wind resource classification systems, this value corresponds to Class 3 wind resource, which is moderate wind energy potential. Sites within this range are generally considered suitable for wind energy development, particularly with modern turbine technologies.

6. Wind Direction Analysis

We looked at wind direction details with a wind rose diagram. It nicely shows that most wind comes from the west and south-west, as seen in Figure 6.1.



This directional pattern indicates a steady prevailing wind regime. Knowing the prevailing wind direction is helpful when designing a wind farm, as it allows us to position turbines to reduce wake effects and maximise energy capture.

7. Temporal Dependence of Wind Speed

We examined how wind speed changes over time using two specialised analysis techniques: the autocorrelation function (ACF) and the partial autocorrelation function (PACF).

The autocorrelation showed that wind speeds tend to stay similar in the short term, meaning what happened in the past few hours influences the current wind speed. This is quite normal for weather patterns and shows how weather systems gradually change.

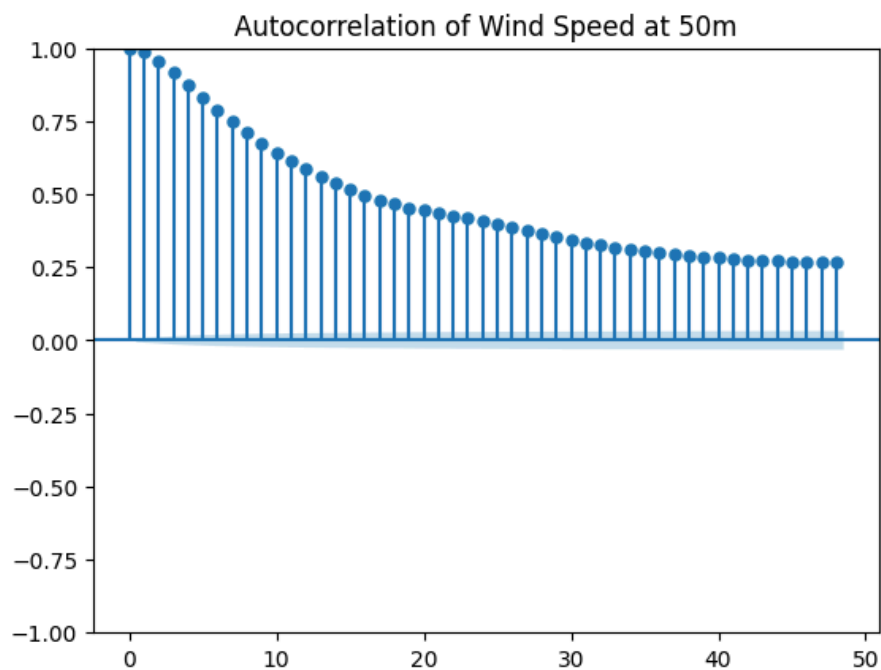


Figure 7.1 Autocorrelation Function

The partial autocorrelation results suggest that simple models like AR(1) or AR(2) could accurately predict future wind speeds. Overall, this means our data is well-suited for developing forecasting methods using statistical or machine-learning approaches.

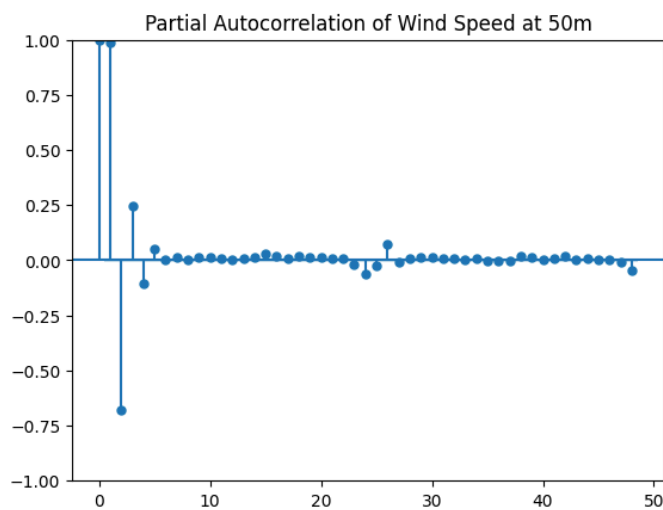


Figure 7.2 Partial autocorrelation of wind speed

8. Wind Speed Duration Curve

A Wind Speed Duration Curve (WSDC) was created by arranging wind speeds from highest to lowest and plotting them against their frequency of occurrence. This curve helps us understand how often different wind speeds occur during the observation period. The analysis revealed that many of the wind speeds fall within a range that's suitable for wind turbines, highlighting the site's good potential for wind energy, as shown in Figure 8.1. Wind speed duration curves are a helpful tool in wind resource assessment, offering valuable insights into how wind turbines might perform and estimating the possible energy they can generate.

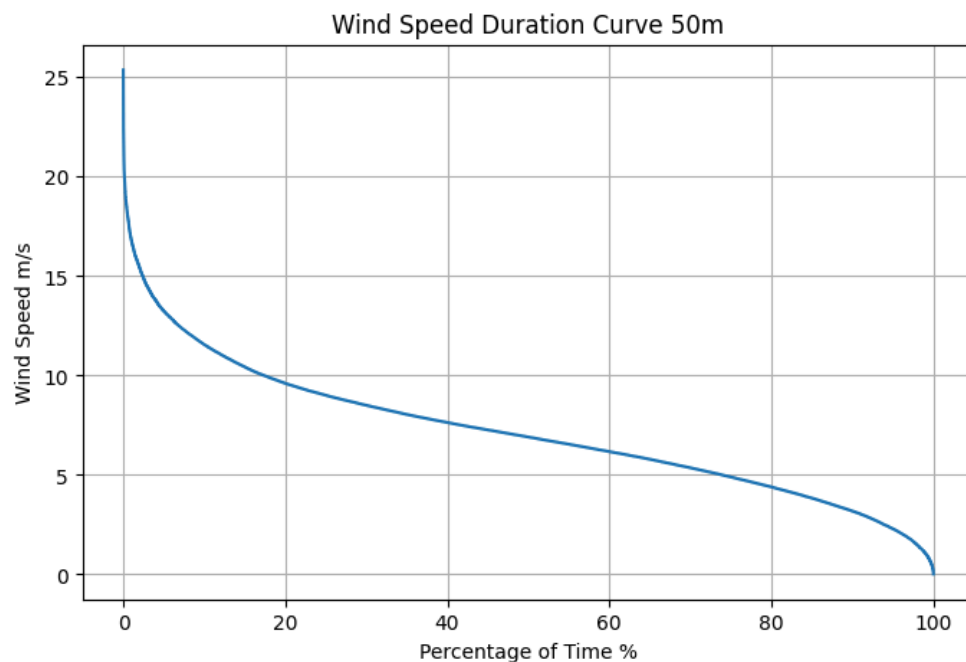


Figure 8.1 Wind speed duration curve

9. Annual Energy Production Estimation

To estimate how much energy a wind turbine could generate at this site, we used a simple power curve for a 2MW turbine based on the wind speed data collected. This model considers a cut-in wind speed of 3 m/s, reaches its rated power at about 12 m/s, and stops operating at a cut-out wind speed of 25 m/s. We calculated the hourly power output for each wind speed observation and added them all up. Since the data covers nearly 10 years, we normalized the total energy to find the average annual energy production (AEP). Overall, this gives us a friendly estimate of the potential yearly electricity a wind turbine could produce here, based on the site's wind conditions.

10. Conclusion

The analysis of the 10-year hourly wind speed dataset shows that this location has great potential for wind energy projects. With an average wind speed of around 7.19 m/s and a wind power density of about 388 W/m², it indicates a decent to good wind resource. The weibull distribution effectively captures the wind speed variations, and the wind rose analysis highlights steady prevailing winds coming from the west and southwest. Additionally, the temporal analysis demonstrates strong autocorrelation, suggesting that wind speed forecasting models could work very well here. Overall, these findings point to a promising site for wind power development, and the dataset offers a reliable basis for further exploration, such as detailed wind farm planning and sophisticated forecasting models.